

FYDP Management & Matchmaking System

Full Website Master Prompt for Antigravity

You are a WORLD-CLASS Full Stack Web Developer and UI/UX Designer.

Your task is to build a COMPLETE, PRODUCTION-READY, FULLY FUNCTIONAL website for:

"FYDP Management & Matchmaking System"

University: UIU (United International University)

This website connects to an already-built MySQL 8.0 database called: `fydp_system`

The website MUST be:

- Professional and modern
 - Fully functional (not a mockup)
 - Role-based (different views per role)
 - Mobile responsive
 - Clean and impressive for university judges
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TECH STACK

Frontend:

- HTML5, CSS3, JavaScript (Vanilla or React)
- Tailwind CSS for styling
- Chart.js for dashboard analytics
- Font Awesome icons

Backend:

- Node.js + Express.js
- MySQL2 package for database connection
- Express Session for authentication
- Bcrypt for password handling
- Multer for file uploads (PDF reports)
- Nodemailer for email notifications

Database:

- Already built MySQL 8.0 (`fydp_system`)
 - Connect using mysql2
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SYSTEM ROLES

The system has 4 roles. Each gets a DIFFERENT dashboard:

1. ADMIN
 2. SUPERVISOR
 3. COURSE_TEACHER
 4. STUDENT
-

PAGE 1 — LOGIN PAGE

Design:

- University logo at top (UIU)
- Clean centered login card
- Email + Password fields
- "Login" button
- Role auto-detected from database
- Redirect to correct dashboard based on role

- Show error for wrong credentials
- Professional dark blue + white color theme

Backend:

- `POST /auth/login`
 - Check email + password_hash (SHA2 256)
 - Create session with user_id, role, full_name
 - Redirect based on role
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DASHBOARD 1 — ADMIN DASHBOARD

Route: `/admin/dashboard`

Stats Cards Row (top):

- Total Students
- Total Supervisors
- Total Active Groups
- FYDP-1 Groups count
- FYDP-2 Groups count
- FYDP-3 Groups count
- Overdue Groups (flagged red)

Charts:

- Bar chart: Groups per FYDP Stage
- Pie chart: Project Domains distribution
- Line chart: Weekly report submissions this month

Tables:

- All Sections with group count
- Recent promotions history
- Recent audit log entries (last 10)

Sidebar Menu:

- Dashboard
 - Manage Users
 - Manage Groups
 - FYDP Stages
 - Promote Groups
 - Import Groups (Bulk)
 - Topic Change History
 - Audit Logs
 - Notifications
-

ADMIN — MANAGE USERS PAGE

Route: `/admin/users`

Features:

- Table of ALL users (Admin, Supervisor, Teacher, Student)
 - Filter by: Role, Department, Batch, Status
 - Search by name or university ID
 - Add new user (form modal)
 - Edit user (form modal)
 - Soft delete user (`is_active = 0`)
 - View student skill profile
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ADMIN — MANAGE GROUPS PAGE

Route: `/admin/groups`

Features:

- Table of ALL groups
 - Filter by: FYDP Stage, Section, Status, Domain, Supervisor
 - See group members
 - Assign/change supervisor
 - Change project title or domain
 - View weekly report status per group
 - Promote group to next stage (calls `sp_promote_fydp_stage`)
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ADMIN — BULK IMPORT PAGE

Route: `/admin/import`

Features:

- Upload CSV file (simulate `ucam_import_staging`)
 - Show preview of CSV data
 - Run import (calls `sp_bulk_import_ucam_groups`)
 - Show import results:
 - Success count
 - Failed rows with error messages (from `import_error_logs`)
 - Download error report
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DASHBOARD 2 — SUPERVISOR DASHBOARD

Route: `/supervisor/dashboard`

Stats Cards:

- Total My Groups
- Pending Reports (needs my approval)
- Approved This Week

- Groups per Stage (FYDP-1, FYDP-2, FYDP-3)

My Groups Table:

| Group Code | Project Title | Stage | Section | Members | Latest Week | Report Status | Action |

Color coding:

-  Green = All approved
-  Yellow = Pending
-  Red = Overdue (7+ days no submission)

Sidebar Menu:

- Dashboard
 - My Groups
 - Pending Approvals
 - Report History
 - Notifications
-

SUPERVISOR — PENDING APPROVALS PAGE

Route: `/supervisor/approvals`

Features:

- List of all PENDING reports across all my groups
- Show: Student Name | Group | Week No | Submitted At | Report Content
- Two action buttons per report:
 -  APPROVE
 -  REQUEST CHANGES
- When clicking APPROVE:
 - Update supervisor_status = APPROVED
 - Set supervisor_signed_at = NOW()

- Trigger fires automatically (escalation check)
 - Show success message
 - When clicking REQUEST CHANGES:
 - Open modal with feedback text box
 - Update status = REJECTED + save feedback
 - Student gets notified
 - After ALL members approved → show “Escalated to Course Teacher ”
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SUPERVISOR — MY GROUPS PAGE

Route: `/supervisor/groups`

Features:

- Cards view of all my groups
 - Each card shows:
 - Group Code + Project Title
 - FYDP Stage badge
 - Section code
 - Member count
 - Progress bar (approved weeks / total weeks)
 - Click to expand → see all members + all weekly reports
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DASHBOARD 3 — COURSE TEACHER DASHBOARD

Route: `/teacher/dashboard`

Stats Cards:

- My Sections
- Groups in My Section

- Escalated Reports Pending Review
- Approved by me this week

Inbox Table:

- All escalated weekly submissions
- Columns: Group | Week | Escalated At | Status | Action
- Action: Mark as Reviewed

Sidebar Menu:

- Dashboard
 - My Inbox
 - Section Groups
 - Notifications
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COURSE TEACHER — INBOX PAGE

Route: `/teacher/inbox`

Features:

- List of all escalated group-week packages
 - Escalated = ALL members in group got supervisor approval for that week
 - Course teacher can: Mark as Reviewed 
 - Filter by: Section, Week No, Status
 - Show which groups are pending review
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DASHBOARD 4 — STUDENT DASHBOARD

Route: `/student/dashboard`

Stats Cards:

- My Group Name

- FYDP Stage
- Current Week
- Reports Submitted
- Reports Approved
- Reports Pending

My Weekly Reports Timeline:

- Visual timeline Week 1 to Week 12
- Each week shows:
 -  Approved
 -  Pending
 -  Changes Requested
 -  Not Submitted
- Click any week to see details + supervisor feedback

Sidebar Menu:

- Dashboard
- My Group
- Submit Report
- My Reports
- Team Matchmaking
- My Skills
- Notifications

STUDENT — SUBMIT WEEKLY REPORT PAGE

Route: `/student/submit-report`

Features:

- Select Week Number (dropdown)

- Report Title (text input)
 - Report Content (rich text area)
 - Upload PDF (optional — multer)
 - Submit button
 - Validation:
 - Cannot submit same week twice (UNIQUE constraint will reject)
 - Show helpful error if duplicate
 - Show success and redirect to My Reports
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STUDENT — MY REPORTS PAGE

Route: `/student/reports`

Features:

- Table of all my submitted reports
 - Columns: Week | Title | Submitted At | Status | Feedback | Action
 - Status color badges:
 -  PENDING
 -  APPROVED
 -  REJECTED
 - If REJECTED: Show supervisor feedback text
 - If REJECTED: Show "Resubmit" button
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STUDENT — TEAM MATCHMAKING PAGE

Route: `/student/matchmaking`

Features:

- Search board of students looking for teams

- Filter by: Skill, Proficiency, Domain Interest, Preferred Role
 - Each student card shows:
 - Name | CGPA | Skills | Domain Interests
 - Preferred Role badge
 - "Send Invitation" button
 - Cannot send to self (CHECK constraint)
 - Cannot send duplicate PENDING invitations (trigger blocks it)
 - Show: My Sent Invitations | My Received Invitations
 - Accept / Reject received invitations
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STUDENT — MY SKILLS PAGE

Route: `/student/skills`

Features:

- View current skills with proficiency level
 - Add new skill:
 - Dropdown of skills from master list
 - Select proficiency (Beginner/Intermediate/Advanced/Expert)
 - Years of experience input
 - Delete skill
 - Add domain interests with interest level (LOW/MEDIUM/HIGH)
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NOTIFICATIONS PAGE (ALL ROLES)

Route: `/notifications`

Features:

- List of all notifications for logged-in user

- Mark as read individually or “Mark All Read”
 - Color coded by type:
 -  Invitation
 -  Approved
 -  Rejected
 -  Escalation
 -  System
 - Real time badge count in header/sidebar
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GLOBAL FEATURES (ALL PAGES)

Header:

- University logo
- System name: “FYDP Management System”
- User name + role badge
- Notification bell with unread count
- Logout button

Sidebar:

- Collapsible
- Active page highlighted
- Role-specific menu items

Color Theme:

- Primary: Dark Blue #1a237e
- Secondary: Light Blue #1565c0
- Accent: Green #2e7d32 (success)
- Warning: Orange #e65100
- Danger: Red #b71c1c

- Background: Light Gray #f5f5f5
 - Cards: White with subtle shadow
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API ROUTES REQUIRED

AUTH:

POST /auth/login
POST /auth/logout
GET /auth/me

ADMIN:

GET /api/admin/stats
GET /api/admin/users
POST /api/admin/users
PUT /api/admin/users/:id
DELETE /api/admin/users/:id
GET /api/admin/groups
POST /api/admin/promote-group
POST /api/admin/import
GET /api/admin/audit-logs
GET /api/admin/topic-history

SUPERVISOR:

GET /api/supervisor/dashboard
GET /api/supervisor/groups
GET /api/supervisor/pending-reports
PUT /api/supervisor/reports/:id/approve
PUT /api/supervisor/reports/:id/reject

COURSE TEACHER:

GET /api/teacher/dashboard
GET /api/teacher/inbox
PUT /api/teacher/inbox/:id/review

STUDENT:

GET /api/student/dashboard
GET /api/student/group
GET /api/student/reports
POST /api/student/reports
GET /api/student/matchmaking
POST /api/student/invitations
PUT /api/student/invitations/:id
GET /api/student/skills

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POST    /api/student/skills
DELETE  /api/student/skills/:id
GET     /api/notifications
PUT     /api/notifications/read-all
```

DATABASE CONNECTION

```
const mysql = require('mysql2/promise');

const db = mysql.createPool({
  host: 'localhost',
  user: 'root',
  password: '',
  database: 'fydp_system',
  waitForConnections: true,
  connectionLimit: 10
});

module.exports = db;
```

FOLDER STRUCTURE

```
fydp-website/
├── server.js
├── package.json
├── .env
├── config/
│   └── db.js
├── routes/
│   ├── auth.js
│   ├── admin.js
│   ├── supervisor.js
│   ├── teacher.js
│   └── student.js
├── middleware/
│   ├── auth.middleware.js
│   └── role.middleware.js
├── public/
│   ├── css/
│   │   └── style.css
│   └── js/
```

```
| | | └─ main.js
| | └─ uploads/
| └─ views/
| | └─ login.html
| | └─ admin/
| | | └─ dashboard.html
| | | └─ users.html
| | | └─ groups.html
| | | └─ import.html
| | └─ supervisor/
| | | └─ dashboard.html
| | | └─ approvals.html
| | | └─ groups.html
| | └─ teacher/
| | | └─ dashboard.html
| | | └─ inbox.html
| | └─ student/
| | | └─ dashboard.html
| | | └─ submit-report.html
| | | └─ reports.html
| | | └─ matchmaking.html
| | | └─ skills.html
```

IMPORTANT RULES

1. Every page MUST be fully functional — no dummy buttons
 2. All forms must connect to real database
 3. Show loading spinners during API calls
 4. Show success/error toast notifications
 5. All tables must have pagination
 6. All data must come from fydp_system database
 7. Session must expire after 2 hours
 8. Protect all routes — redirect to login if not authenticated
 9. Role protection — student cannot access admin routes
 10. Mobile responsive — must work on phone
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OUTPUT ORDER

1. package.json
 2. server.js
 3. config/db.js
 4. middleware files
 5. All route files
 6. All HTML pages (with CSS inline or linked)
 7. Main CSS file
 8. Main JS file
 9. README with setup instructions
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FINAL EXPECTATION

This website MUST look and feel like a REAL university management system. Clean. Professional. Fast. Functional. Every button works. Every form saves to database. Every role sees only what they should see. This is a competition project — make it IMPRESSIVE.

FYDP Management System — Built to Win | UIU DBMS Showcase